CODE APPENDIX

Libraries used.

```r
# Core data manipulation and visualization (loads dplyr, ggplot2, tidyr, etc.)
library(tidyverse)

# Modeling, diagnostics, and tidying
library(car)
library(MASS)
library(faraway)
library(lmtest)
library(broom)

# Formatting and reporting
library(knitr)
library(kableExtra)
library(bookdown)
library(ggcorrplot)
```

Helper Functions

```r
# Reusable function to build standard vertical regression statistics tables
create_model_summary <- function(model_object, table_caption) {
  g_stats <- glance(model_object)

  stat_df <- data.frame(
    " " = c("F - Statistic",
            "Residual Standard Error",
            "Multiple R - Squared",
            "Adjusted R - Squared"),
    "Value" = c(
      round(g_stats$statistic, 4),
      round(g_stats$sigma, 4),
      round(g_stats$r.squared, 4),
      round(g_stats$adj.r.squared, 4)
    ),
    "DF" = c(
      paste0(g_stats$df - 1, " and ", g_stats$df.residual),
      as.character(g_stats$df.residual),
      NA,
      NA
```

```r
    ),
    "Pr(>|t|)" = c(
      format.pval(g_stats$p.value, digits = 4),
      NA,
      NA,
      NA
    ),
    check.names = FALSE
  )

  options(knitr.kable.NA = ' ')

  kable(
    stat_df,
    caption = table_caption,
    align = c("l", "c", "c", "c"),
    format = "latex",
    booktabs = TRUE
  )
}
```

Methodology

$$\widehat{quality} = \hat{\beta}_0 + \hat{\beta}_1 gender + \hat{\beta}_2 numYears + \hat{\beta}_3 numRaters$$

$$+\hat{\beta}_4 numCourses + \hat{\beta}_5 pepper + \hat{\beta}_6 discipline + \hat{\beta}_7 dept$$

$$+\hat{\beta}_8 helpfulness + \hat{\beta}_9 clarity + \hat{\beta}_{10} easiness$$

$$+\hat{\beta}_{11} raterInterest + \hat{\beta}_{12} sdQuality + \hat{\beta}_{13} sdHelpfulness$$

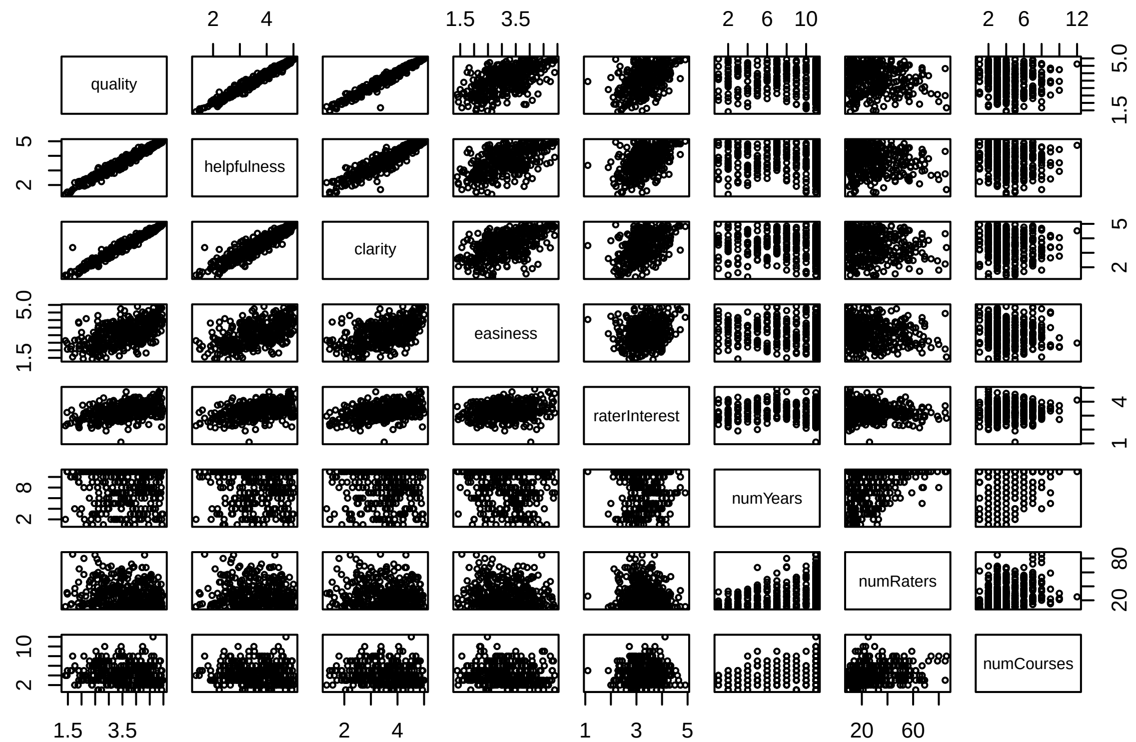
$$+\hat{\beta}_{14} sdClarity + \hat{\beta}_{15} sdEasiness + \hat{\beta}_{16} sdRaterInterest$$

Load Data

```r
library(alr4)
data("Rateprof")
```

Preliminary Exploratory Analysis: Pairs Plot

```r
par(cex = 0.8)
par(mar = c(2,2,2,2))
pairs(quality ~ helpfulness + clarity + easiness + raterInterest +  numYears
      + numRaters + numCourses, data = Rateprof, cex = 0.55)
```



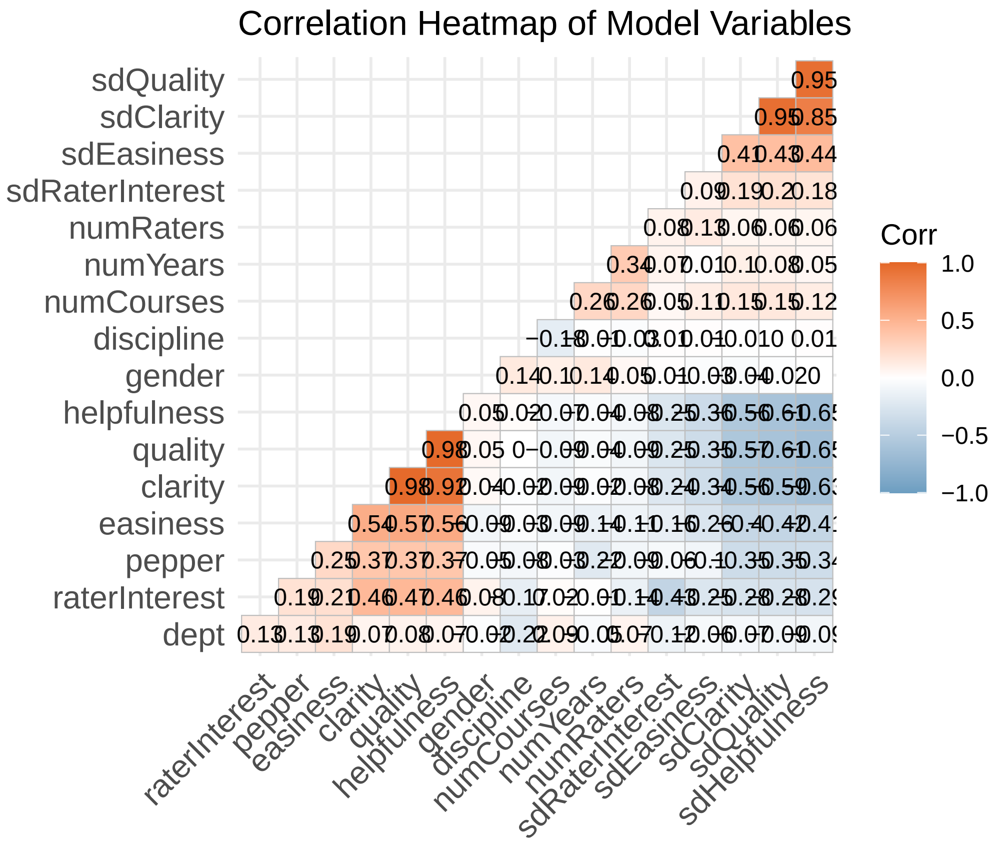
Summary Statistics: correlation matrix

```r
#Create data frame to convert variables to numeric to use for cor()
num_df <- Rateprof
#Convert to numeric
num_df <- num_df %>%
  mutate(across(where(is.factor), as.numeric))

#Compute correlation matrix
cor_matrix <- cor(num_df)

#Plot an elegant heatmap showing only the lower triangle
ggcorrplot(cor_matrix,
           hc.order = TRUE,
           type = "lower",
           lab = TRUE,           # Adds text labels inside squares
```

```
            lab_size = 3,         # Perfect size for many variables
            method = "square",
            colors = c("#6D9EC1", "white", "#E46726"), # Clean professional gradient
            title = "Correlation Heatmap of Model Variables",
            ggtheme = theme_minimal())
```




```
#Compute correlation matrix
cor_tb <- (round(cor(num_df), 2))

options(knitr.kable.NA = ' ')
tb1_cor <- kable(
  cor_tb,
  caption = "Correalation Matrix Table",
  align = c("l", "c", "c", "c"),
  digits = 4,
  format = "latex",
  booktabs = TRUE
) %>%
  kable_styling(latex_options = "scale_down")

tb1_cor
```

Table 1: Correalation Matrix Table

	gender	numYears	numRaters	numCourses	pepper	discipline	dept	quality	helpfulness	clarity	easiness	raterInterest	sdQuality	sdHelpfulness	sdClarity	sdEasiness	sdRaterInterest
gender	1.00	0.14	0.05	0.10	-0.05	0.14	-0.02	0.05	0.05	0.04	-0.09	0.08	-0.02	0.00	-0.04	-0.03	0.01
numYears	0.14	1.00	0.34	0.26	-0.22	-0.01	-0.05	-0.04	-0.04	-0.02	-0.14	-0.01	0.08	0.05	0.10	0.01	0.07
numRaters	0.05	0.34	1.00	0.26	-0.09	-0.03	0.07	-0.09	-0.08	-0.08	-0.11	-0.14	0.06	0.06	0.06	0.13	0.08
numCourses	0.10	0.26	0.26	1.00	-0.03	-0.18	0.09	-0.09	-0.07	-0.09	-0.09	0.02	0.15	0.12	0.15	0.11	0.05
pepper	-0.05	-0.22	-0.09	-0.03	1.00	-0.08	0.13	0.37	0.37	0.37	0.25	0.19	-0.35	-0.34	-0.35	-0.10	-0.06
discipline	0.14	-0.01	-0.03	-0.18	-0.08	1.00	-0.22	0.00	0.02	-0.02	-0.03	-0.17	0.00	0.01	-0.01	0.01	0.01
dept	-0.02	-0.05	0.07	0.09	0.13	-0.22	1.00	0.08	0.07	0.07	0.19	0.13	-0.09	-0.09	-0.07	-0.06	-0.12
quality	0.05	-0.04	-0.09	-0.09	0.37	0.00	0.08	1.00	0.98	0.98	0.57	0.47	-0.61	-0.65	-0.57	-0.35	-0.25
helpfulness	0.05	-0.04	-0.08	-0.07	0.37	0.02	0.07	0.98	1.00	0.92	0.56	0.46	-0.61	-0.65	-0.56	-0.36	-0.25
clarity	0.04	-0.02	-0.08	-0.09	0.37	-0.02	0.07	0.98	0.92	1.00	0.54	0.46	-0.59	-0.63	-0.56	-0.34	-0.24
easiness	-0.09	-0.14	-0.11	-0.09	0.25	-0.03	0.19	0.57	0.56	0.54	1.00	0.21	-0.42	-0.41	-0.40	-0.26	-0.16
raterInterest	0.08	-0.01	-0.14	0.02	0.19	-0.17	0.13	0.47	0.46	0.46	0.21	1.00	-0.28	-0.29	-0.28	-0.25	-0.43
sdQuality	-0.02	0.08	0.06	0.15	-0.35	0.00	-0.09	-0.61	-0.61	-0.59	-0.42	-0.28	1.00	0.95	0.95	0.43	0.20
sdHelpfulness	0.00	0.05	0.06	0.12	-0.34	0.01	-0.09	-0.65	-0.65	-0.63	-0.41	-0.29	0.95	1.00	0.85	0.44	0.18
sdClarity	-0.04	0.10	0.06	0.15	-0.35	-0.01	-0.07	-0.57	-0.56	-0.56	-0.40	-0.28	0.95	0.85	1.00	0.41	0.19
sdEasiness	-0.03	0.01	0.13	0.11	-0.10	0.01	-0.06	-0.35	-0.36	-0.34	-0.26	-0.25	0.43	0.44	0.41	1.00	0.09
sdRaterInterest	0.01	0.07	0.08	0.05	-0.06	0.01	-0.12	-0.25	-0.25	-0.24	-0.16	-0.43	0.20	0.18	0.19	0.09	1.00

Table 2: Regression Statistics: Full Additive Model I

	Value	DF	Pr(>|t|)
F - Statistic	1893.8518	60 and 304	< 2.2e-16
Residual Standard Error	0.0470	304	
Multiple R - Squared	0.9974		
Adjusted R - Squared	0.9968		

Fit the Model: Full Additive Model

```r
#Full additive model
lm1 <- lm(quality ~ gender + numYears + numRaters
    + numCourses + pepper + discipline + dept
    + helpfulness + clarity + easiness
    + raterInterest + sdQuality + sdHelpfulness
    + sdClarity + sdEasiness
    + sdRaterInterest, Rateprof)
```

Summary output for full additive model.

```r
#Model I Summary Statistics Table
# For Model I
create_model_summary(lm1, "Regression Statistics: Full Additive Model I")
```

Check Assumptions

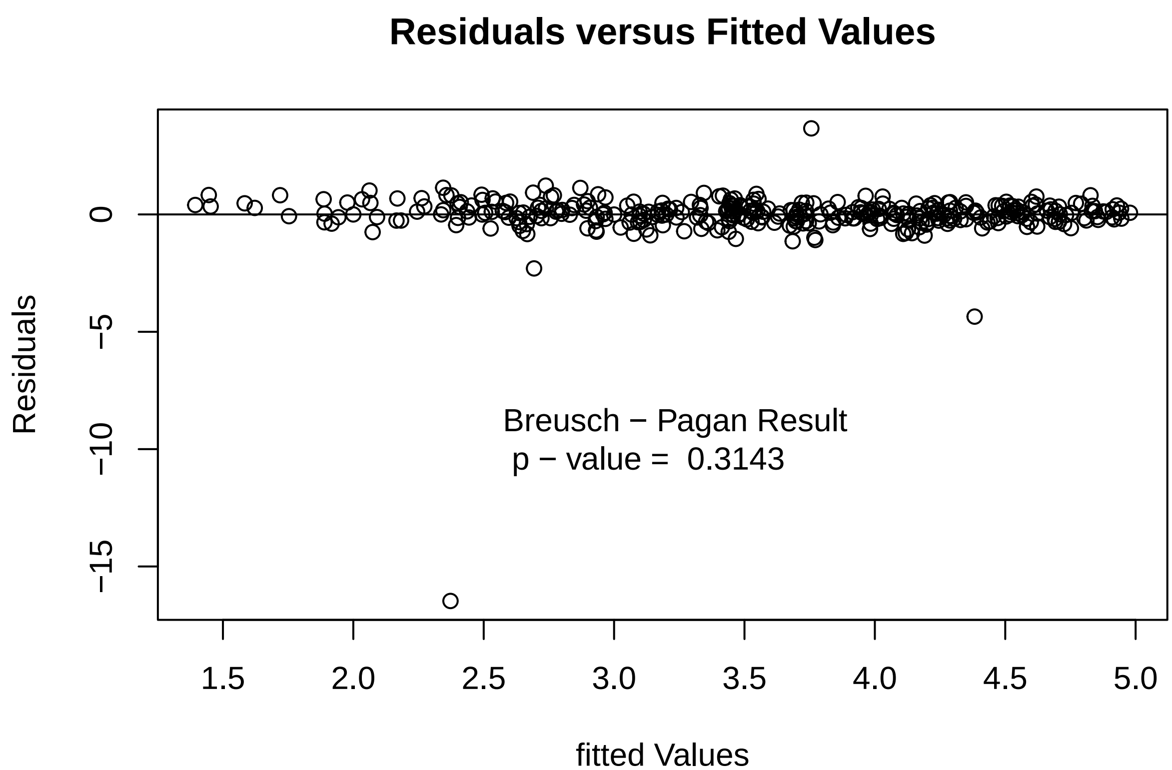
Constant Variance: residuals versus fitted values plot with Breusch - Pagan Result

```r
#Residuals versus fitted plot
plot(fitted(lm1), rstandard(lm1),
     main = "Residuals versus Fitted Values",
```

```r
     xlab = "fitted Values",
     ylab = "Residuals")
abline(h = 0)

#Breusch - Pagan Test
bp1 <- bptest(lm1)

#Add Breusch - Pagan results to residuals versus fitted plot
text(x = 2.5, y = -10,
     labels = paste("Breusch - Pagan Result",
                    "\n p - value = ", round(bp1$p.value, 4)),
     pos = 4, col = "black")
```

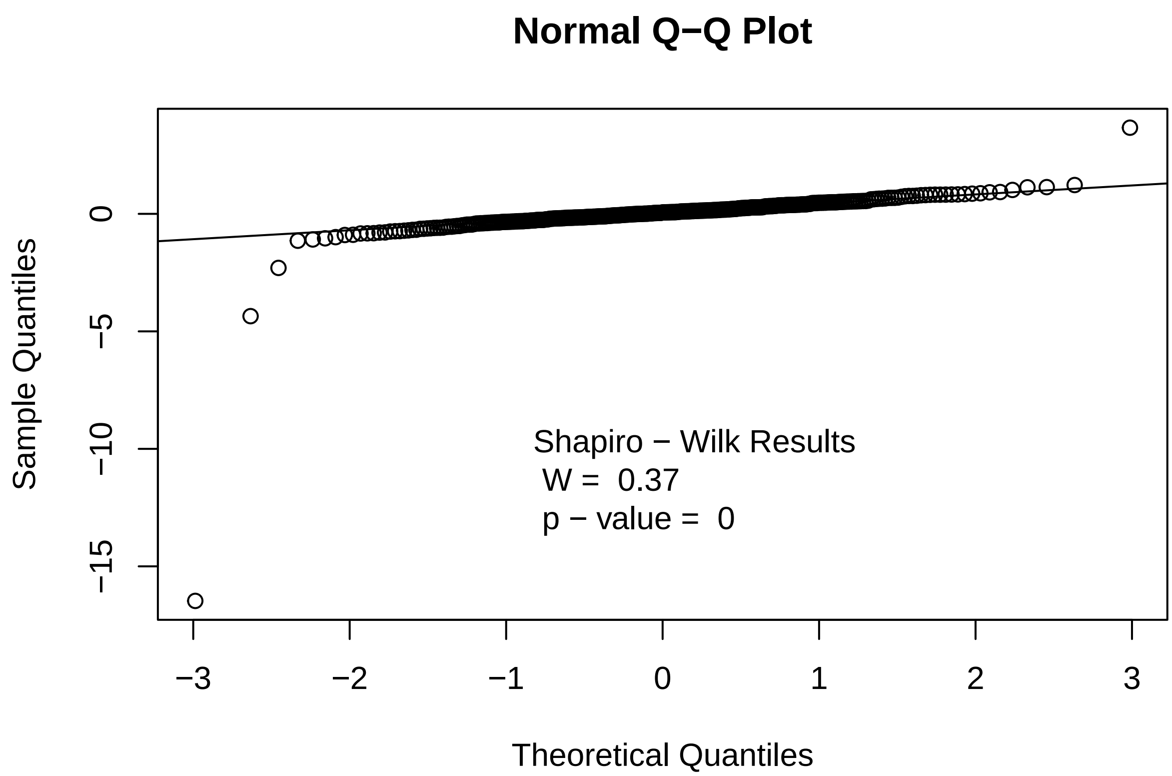


Normality: Normal Q - Q Plot with Shapiro - Wilk Result

```r
#Normal Q - Q plot
qqnorm(rstandard(lm1))
qqline(rstandard(lm1))

#Shapiro Wilk test
shap1 <- shapiro.test(resid(lm1))
```

```
#Add Shapiro results to Normal Q - Q plot
text(x = -0.95, y = -12,
     labels = paste("Shapiro - Wilk Results",
                    "\n W = ", round(shap1$statistic, 4),
                    "\n p - value = ", round(shap1$p.value, 4)),
      pos = 4, col = "black")
```



Fit the model: Use the stepwise function on the full additive model.

```
#Recall:
lm1 <- lm(quality ~ gender + numYears + numRaters
    + numCourses + pepper + discipline + dept +
    + helpfulness + clarity + easiness +  raterInterest
    + sdQuality + sdHelpfulness + sdClarity + sdEasiness
    + sdRaterInterest, Rateprof)

#Step() with full additive model
lm2 <- step(lm1, trace = 0)
```

Table 3: Regression Statistics: Stepwise Additive Model II

	Value	DF	Pr(>|t|)
F - Statistic	25900.9196	4 and 360	< 2.2e-16
Residual Standard Error	0.0444	360	
Multiple R - Squared	0.9972		
Adjusted R - Squared	0.9972		

Summary output for the stepwise model from the full additive model.

```r
#Model II Summary Statistics
create_model_summary(lm2, "Regression Statistics: Stepwise Additive Model II")
```

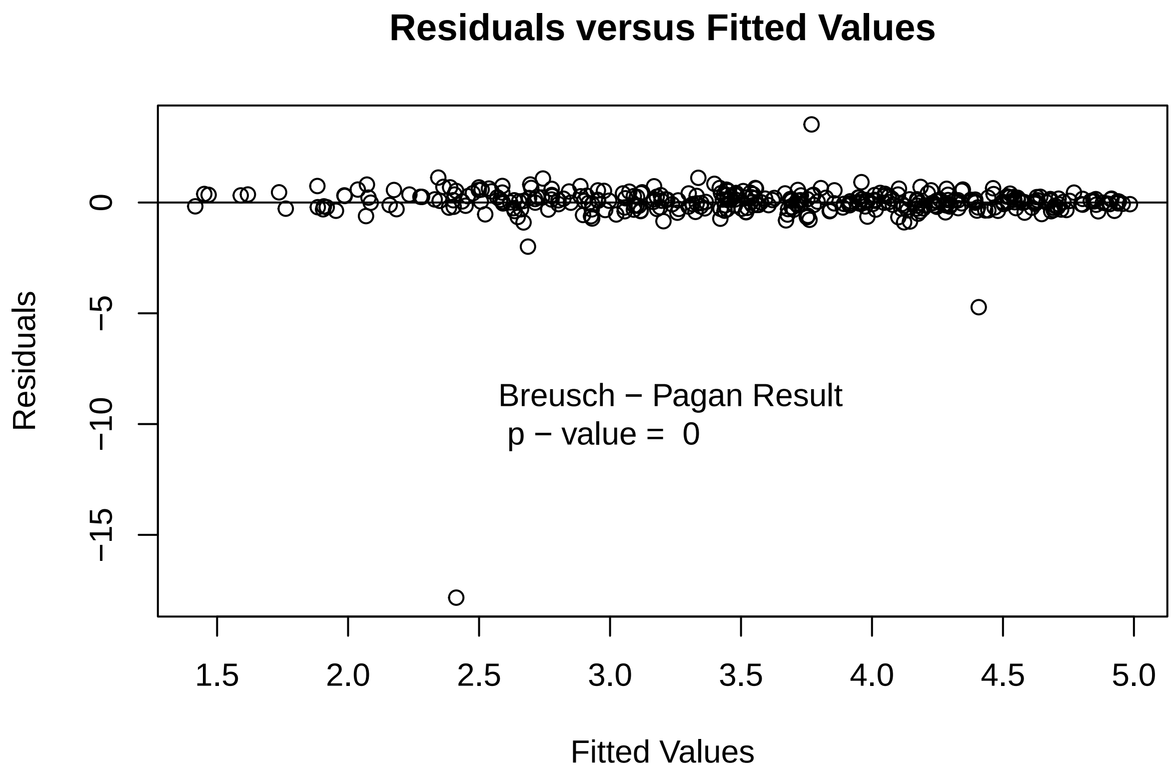
Check Assumptions

Constant Variance: residuals versus fitted values plot along with Breusch - Pagan test

```r
#Residuals versus Fitted plot
plot(fitted(lm2), rstandard(lm2),
     main = "Residuals versus Fitted Values",
     xlab = "Fitted Values",
     ylab = "Residuals")
abline(h = 0)


#Breusch - Pagan Results
bp2 <- bptest(lm2)


#Add Breusch - Pagan results to residual versus fitted plot
text(x = 2.5, y = -10,
     labels = paste("Breusch - Pagan Result",
                    "\n p - value = ", round(bp2$p.value, 4)),
     pos = 4, col = "black")
```

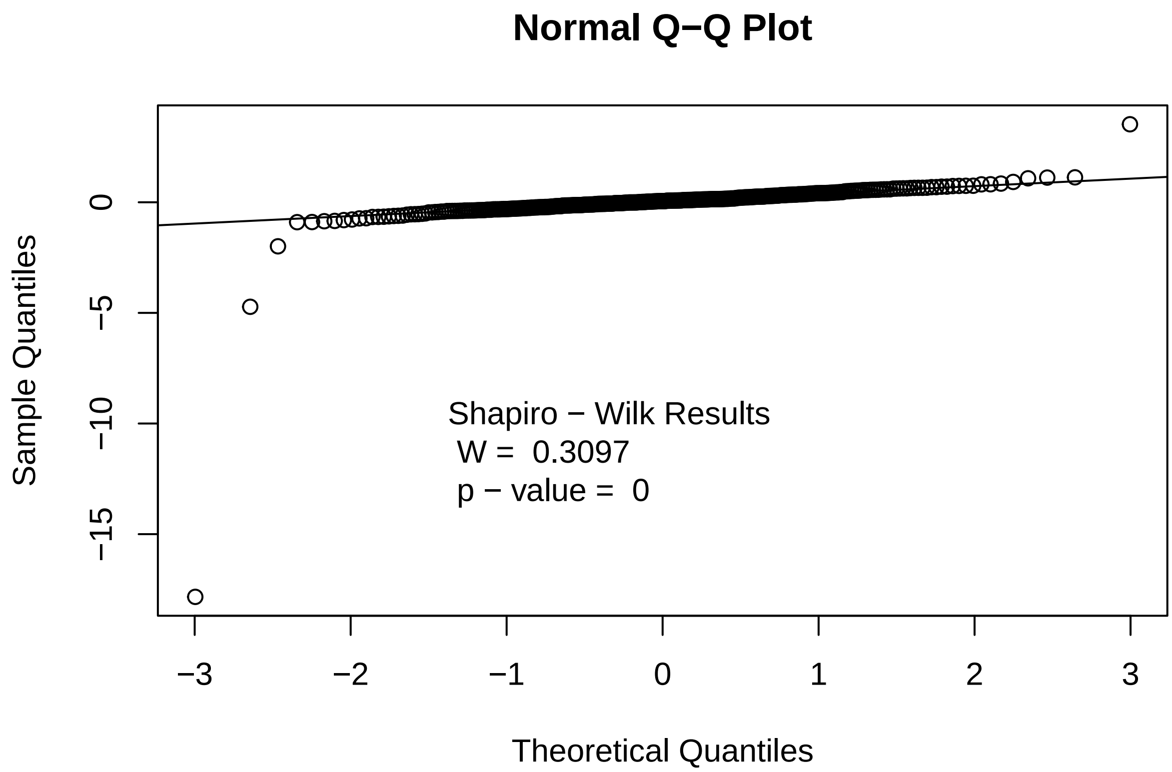


Normality: Normal Q - Q Plot with Shapiro - Wilk Test

```r
#Normal Q - Q plot
qqnorm(rstandard(lm2))
qqline(rstandard(lm2))

#Shapiro - Wilk Test
shap2 <- shapiro.test(resid(lm2))

#Add Shapiro results to Normal Q - Q plot
text(x = -1.5, y = -12,
     labels = paste("Shapiro - Wilk Results",
                    "\n W = ", round(shap2$statistic, 4),
                    "\n p - value = ", round(shap2$p.value, 4)),
      pos = 4, col = "black")
```



Fit the model: Interactions model

Remove helpfulness and clarity due to potential strong linear relationship between the two predictors and the response variable quality assessed by the pairs plot.

```
#Interactions model introduced
#Full model -helpfulness, -clarity

lm3 <- lm(quality ~  discipline + dept +
      (gender + pepper + easiness +
      raterInterest + numYears + numCourses
      + numRaters + sdHelpfulness + sdClarity +
      sdEasiness + sdRaterInterest)^2, Rateprof)
```

Table 4: Regression Statistics: Stepwise Interactions Model III

	Value	DF	Pr(>|t|)
F - Statistic	7.2882	112 and 252	< 2.2e-16
Residual Standard Error	0.4879	252	
Multiple R - Squared	0.7657		
Adjusted R - Squared	0.6606		

Summary output for the interactions model.

```
#Model III Summary Statistics
create_model_summary(lm3, "Regression Statistics: Stepwise Interactions Model III")
```

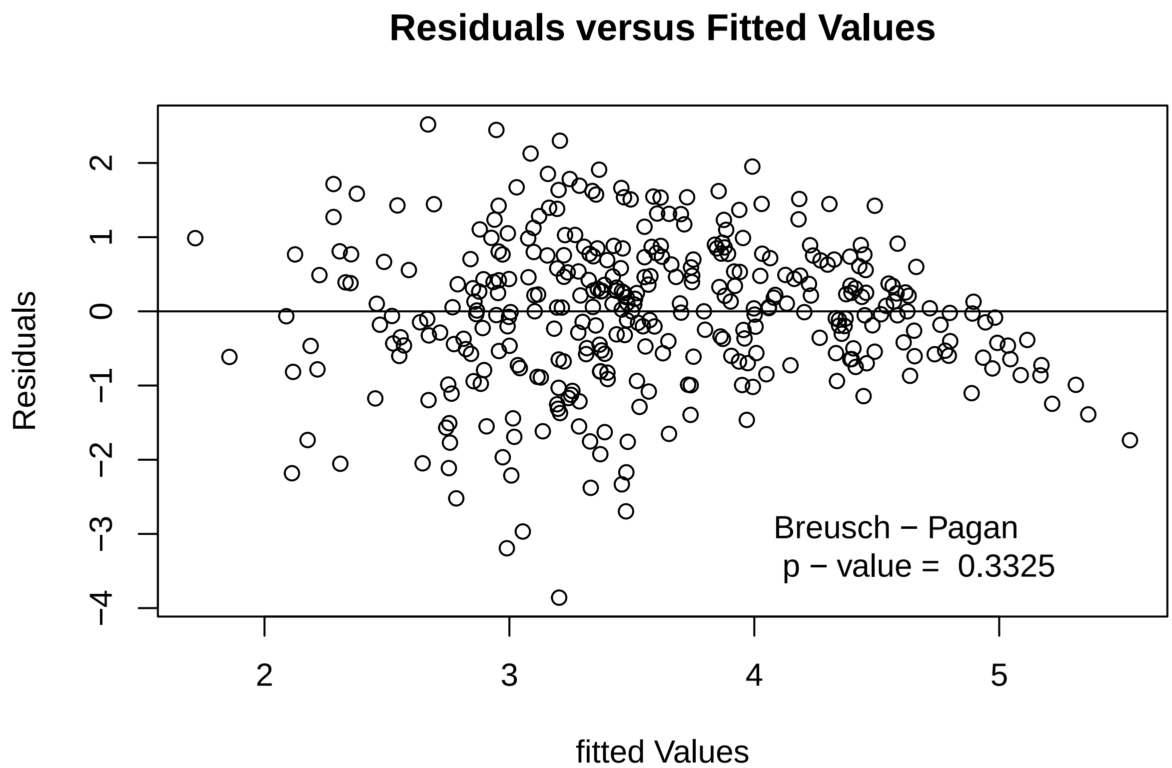
Check Assumptions

Constant Variance: residuals versus fitted values plot with Breusch - Pagan Results

```
#Check lm3 assumptions
plot(fitted(lm3), rstandard(lm3),
     main = "Residuals versus Fitted Values",
     xlab = "fitted Values",
     ylab = "Residuals")
abline(h = 0)

#Breusch - Pagan Test
bp3 <- bptest(lm3)


#Add results to Residuals versus fitted values plot
text(x = 4, y = -3.3,
     labels = paste("Breusch - Pagan",
                    "\n p - value = ", round(bp3$p.value, 4)),
     pos = 4, col = "black")
```

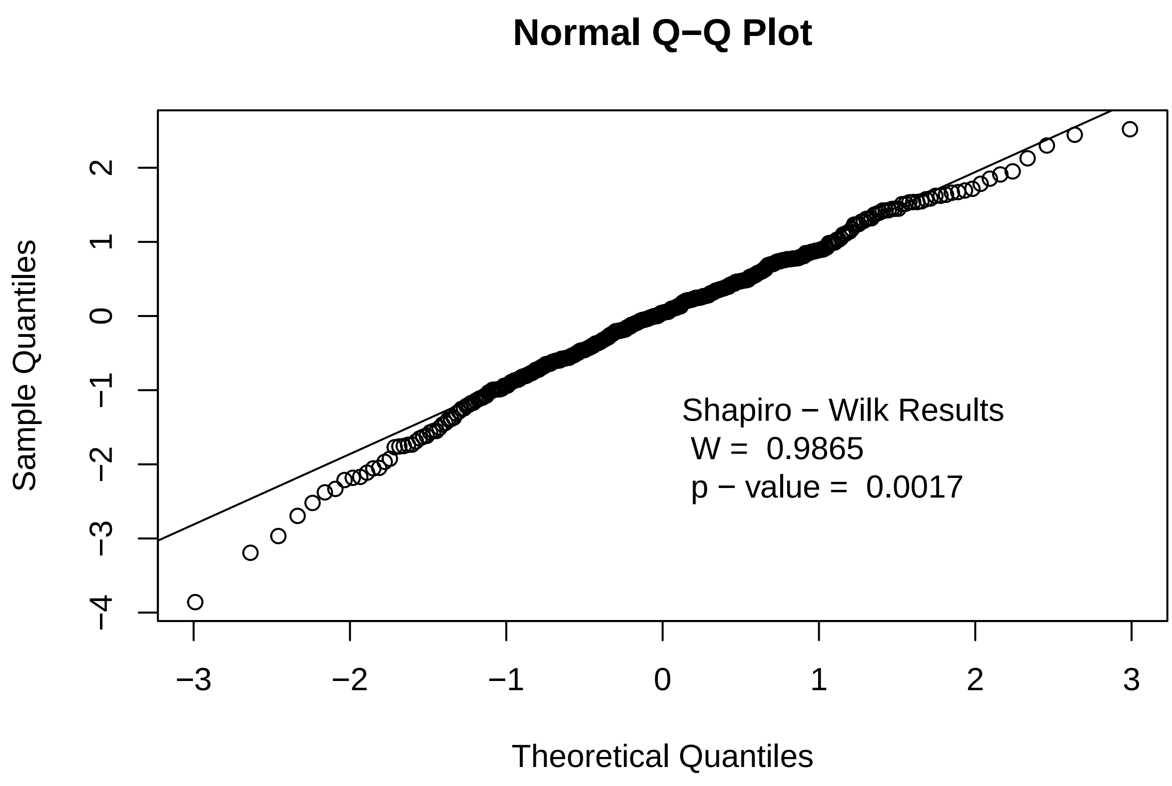

Residuals versus Fitted Values
Residuals
2
1
0
−1
−2
−3
−4
2
3
4
5
fitted Values
Breusch − Pagan
p − value = 0.3325

Normality: Normal Q - Q plot.

```
qqnorm(rstandard(lm3))
qqline(rstandard(lm3))

#Shapiro - Wilk Test
shap3 <- shapiro.test(resid(lm3))

#Add results to Normal Q - Q Plot
text(x = 0, y = -2,
     labels = paste("Shapiro - Wilk Results",
                    "\n W = ", round(shap3$statistic, 4),
                    "\n p - value = ", round(shap3$p.value, 4)),
      pos = 4, col = "black")
```



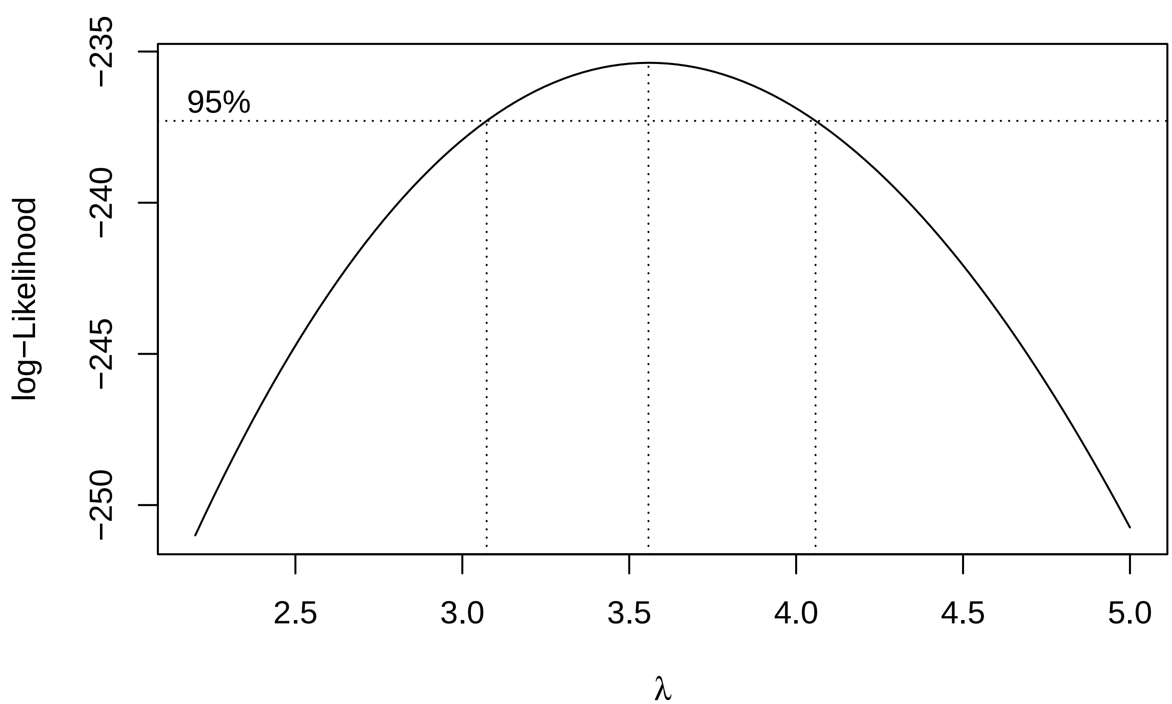
Transformed Model

The assumptions are still violated, therefore check for transformations.

```r
#Try transformation with interaction model
summary(powerTransform(lm3))
```

```
## bcPower Transformation to Normality
##    Est Power Rounded Pwr Wald Lwr Bnd Wald Upr Bnd
## Y1    3.5598        3.56       3.0671       4.0525
##
## Likelihood ratio test that transformation parameter is equal to 0
##  (log transformation)
##                             LRT df       pval
## LR test, lambda = (0) 241.7316  1 < 2.22e-16
##
## Likelihood ratio test that no transformation is needed
##                             LRT df       pval
## LR test, lambda = (1) 117.9941  1 < 2.22e-16
```

```r
boxcox(lm3, lambda = seq(2.2, 5.0, by = 0.05))
```



Results

Final Model: Stepwise Transformed model

Model has been transformed and the stepwise function has been applied to the transformed model.

```
#Transformed model
lm4 <- lm((quality^3.55) ~  discipline + dept +
    (gender + pepper + easiness +  raterInterest
    + numYears + numCourses  + numRaters + sdQuality
    + sdHelpfulness + sdClarity + sdEasiness
    + sdRaterInterest)^2, Rateprof)

#Apply step function
lm5 <- step(lm4, trace = 0)
```

Summary output for final model.

```
#Tidy the model coefficients dynamically
tidy_coef <- tidy(lm5) |>
  mutate(
    across(where(is.numeric), ~ round(., 4)),
    # Add standard scientific significance indicator strings
    Significance = case_when(
      p.value < 0.001 ~ "***",
      p.value < 0.01  ~ "**",
      p.value < 0.05  ~ "*",
      p.value < 0.1   ~ ".",
      TRUE            ~ ""
    )
  )

#Build a beautiful, structured table
kable(
  tidy_coef,
  col.names = c("Predictor", "Estimate", "Std. Error", "t statistic", "Pr(>|t|)", ""),
  caption = "Regression Coefficients: Interactions Model IV",
  format = "latex",
  booktabs = TRUE
) |>
  # Highlight rows that are statistically significant (p < 0.05) to draw the eye
  row_spec(which(tidy_coef$p.value < 0.05), bold = TRUE, background = "#F9F9F9") |>
  # Group the rows into clean structural sections
  pack_rows("Main Effects", 1, 13) |>
  pack_rows("Interaction Terms", 14, nrow(tidy_coef))
```

```r
#Create table
sum_lm5 <- summary(lm5)

#Create data frame for regression coefficients to make table
#Pull coefficients from summary output
lm5_df <- as.data.frame(summary(lm5)$coefficients)

tb5_lm5 <- kable(
  lm5_df,
  caption = "Regression Coefficients: Interactions Model IV",
  format = "latex",
  booktabs = TRUE,
  digits = 4)
tb5_lm5
```

Continued summary output for the final model.

```r
#Model IV Summary Statistics
create_model_summary(lm5, "Regression Statistics: Transformed Model IV")
```

Check Assumptions

Constant Variance: residuals versus fitted values plot with Breusch - Pagan Test

```r
#Check lm5 assumptions
#Residuals versus fitted plot
plot(fitted(lm5), rstandard(lm5),
     main = "Residuals versus Fitted Values",
     xlab = "fitted Values",
     ylab = "Residuals")
abline(h = 0)

#Breusch - Pagan Test
bp5 <- bptest(lm5)

#Add Breusch - Pagan results to Residuals versus fitted plot
text(x = 140, y = -3,
     labels = paste("Breusch - Pagan Result",
                    "\n p - value = ", round(bp5$p.value, 4)),
     pos = 4, col = "black")
```

Table 5: Regression Coefficients: Interactions Model IV

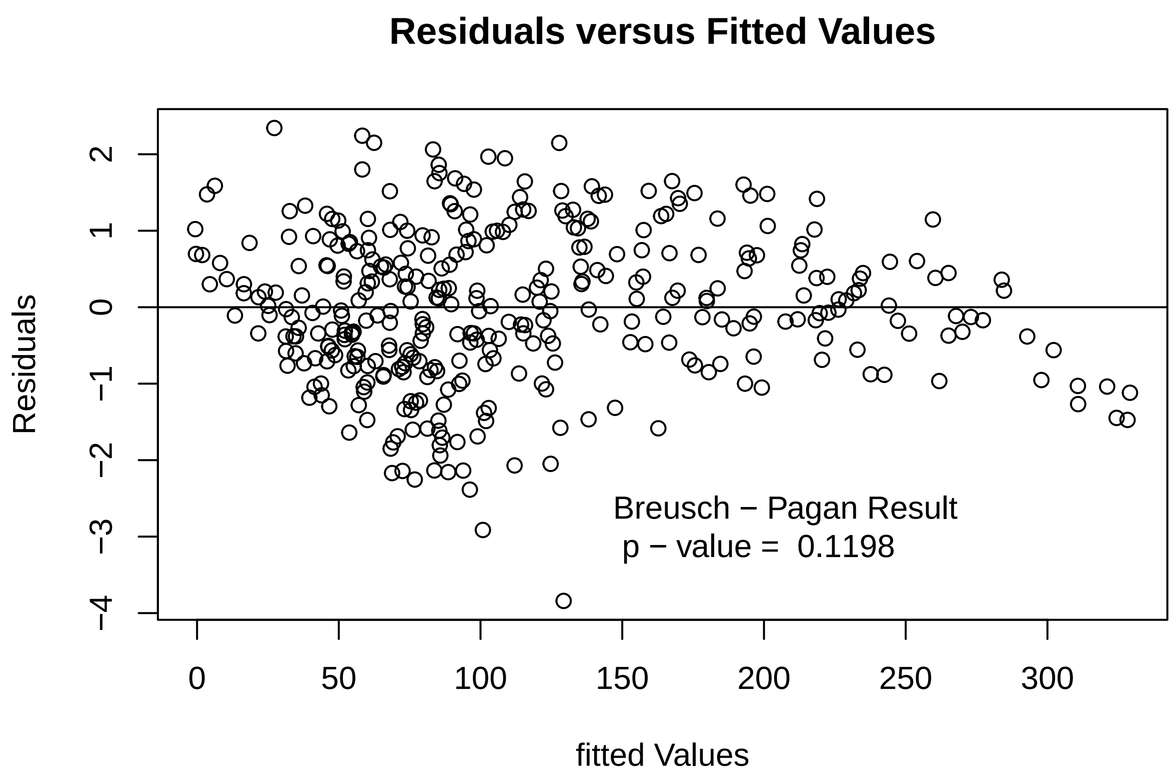
Predictor	Estimate	Std. Error	t statistic	Pr(>|t|)	
Main Effects					
(Intercept)	**-635.4418**	**142.0092**	**-4.4747**	**0.0000**	***
gendermale	1.9668	13.3157	0.1477	0.8827	
pepperyes	**273.6179**	**84.0483**	**3.2555**	**0.0012**	**
easiness	**133.1880**	**23.9593**	**5.5589**	**0.0000**	***
raterInterest	**160.7527**	**26.0077**	**6.1810**	**0.0000**	***
numYears	**11.1823**	**4.9196**	**2.2730**	**0.0237**	*
numCourses	**-29.7393**	**11.0907**	**-2.6815**	**0.0077**	**
numRaters	**4.1821**	**0.9204**	**4.5438**	**0.0000**	***
sdQuality	**696.7181**	**143.1248**	**4.8679**	**0.0000**	***
sdHelpfulness	**-313.9197**	**121.2182**	**-2.5897**	**0.0100**	*
sdClarity	-107.6537	93.1659	-1.1555	0.2487	
sdEasiness	**91.5367**	**40.0222**	**2.2872**	**0.0228**	*
sdRaterInterest	57.6566	35.9237	1.6050	0.1095	
Interaction Terms					
gendermale:sdQuality	-71.4537	38.2115	-1.8700	0.0624	.
gendermale:sdClarity	70.5870	39.7194	1.7771	0.0765	.
pepperyes:raterInterest	**-55.8436**	**16.5366**	**-3.3770**	**0.0008**	***
pepperyes:sdQuality	-80.3646	49.4553	-1.6250	0.1051	
pepperyes:sdClarity	104.6555	54.3565	1.9254	0.0550	.
pepperyes:sdRaterInterest	**-63.1330**	**28.0364**	**-2.2518**	**0.0250**	*
easiness:raterInterest	**-22.1231**	**4.9690**	**-4.4522**	**0.0000**	***
easiness:numCourses	2.5403	1.3805	1.8402	0.0666	.
easiness:numRaters	**-0.4260**	**0.1595**	**-2.6705**	**0.0079**	**
easiness:sdClarity	**-31.9363**	**10.5086**	**-3.0390**	**0.0026**	**
raterInterest:numYears	-2.1359	1.1757	-1.8167	0.0702	.
raterInterest:numCourses	**4.1844**	**2.0977**	**1.9948**	**0.0469**	*
raterInterest:sdQuality	**-135.5478**	**32.1890**	**-4.2110**	**0.0000**	***
raterInterest:sdHelpfulness	**73.1120**	**29.2719**	**2.4977**	**0.0130**	*
numYears:sdClarity	-2.8640	2.0005	-1.4316	0.1532	
numCourses:numRaters	-0.1073	0.0590	-1.8186	0.0699	.
numCourses:sdQuality	**36.3869**	**10.7742**	**3.3772**	**0.0008**	***
numCourses:sdHelpfulness	**-24.1331**	**9.5501**	**-2.5270**	**0.0120**	*
numRaters:sdQuality	**-2.1956**	**0.4974**	**-4.4142**	**0.0000**	***
sdQuality:sdHelpfulness	-91.9033	51.8711	-1.7718	0.0774	.
sdHelpfulness:sdClarity	**119.6386**	**59.5297**	**2.0097**	**0.0453**	*
sdHelpfulness:sdRaterInterest	-48.0130	27.3997	-1.7523	0.0806	.
sdClarity:sdEasiness	**-73.5619**	**33.3979**	**-2.2026**	**0.0283**	*

Table 6: Regression Coefficients: Interactions Model IV

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	-635.4418	142.0092	-4.4747	0.0000
gendermale	1.9668	13.3157	0.1477	0.8827
pepperyes	273.6179	84.0483	3.2555	0.0012
easiness	133.1880	23.9593	5.5589	0.0000
raterInterest	160.7527	26.0077	6.1810	0.0000
numYears	11.1823	4.9196	2.2730	0.0237
numCourses	-29.7393	11.0907	-2.6815	0.0077
numRaters	4.1821	0.9204	4.5438	0.0000
sdQuality	696.7181	143.1248	4.8679	0.0000
sdHelpfulness	-313.9197	121.2182	-2.5897	0.0100
sdClarity	-107.6537	93.1659	-1.1555	0.2487
sdEasiness	91.5367	40.0222	2.2872	0.0228
sdRaterInterest	57.6566	35.9237	1.6050	0.1095
gendermale:sdQuality	-71.4537	38.2115	-1.8700	0.0624
gendermale:sdClarity	70.5870	39.7194	1.7771	0.0765
pepperyes:raterInterest	-55.8436	16.5366	-3.3770	0.0008
pepperyes:sdQuality	-80.3646	49.4553	-1.6250	0.1051
pepperyes:sdClarity	104.6555	54.3565	1.9254	0.0550
pepperyes:sdRaterInterest	-63.1330	28.0364	-2.2518	0.0250
easiness:raterInterest	-22.1231	4.9690	-4.4522	0.0000
easiness:numCourses	2.5403	1.3805	1.8402	0.0666
easiness:numRaters	-0.4260	0.1595	-2.6705	0.0079
easiness:sdClarity	-31.9363	10.5086	-3.0390	0.0026
raterInterest:numYears	-2.1359	1.1757	-1.8167	0.0702
raterInterest:numCourses	4.1844	2.0977	1.9948	0.0469
raterInterest:sdQuality	-135.5478	32.1890	-4.2110	0.0000
raterInterest:sdHelpfulness	73.1120	29.2719	2.4977	0.0130
numYears:sdClarity	-2.8640	2.0005	-1.4316	0.1532
numCourses:numRaters	-0.1073	0.0590	-1.8186	0.0699
numCourses:sdQuality	36.3869	10.7742	3.3772	0.0008
numCourses:sdHelpfulness	-24.1331	9.5501	-2.5270	0.0120
numRaters:sdQuality	-2.1956	0.4974	-4.4142	0.0000
sdQuality:sdHelpfulness	-91.9033	51.8711	-1.7718	0.0774
sdHelpfulness:sdClarity	119.6386	59.5297	2.0097	0.0453
sdHelpfulness:sdRaterInterest	-48.0130	27.3997	-1.7523	0.0806
sdClarity:sdEasiness	-73.5619	33.3979	-2.2026	0.0283

Table 7: Regression Statistics: Transformed Model IV

	Value	DF	Pr(>|t|)
F - Statistic	48.2083	34 and 330	< 2.2e-16
Residual Standard Error	33.2921	330	
Multiple R - Squared	0.8364		
Adjusted R - Squared	0.8191		



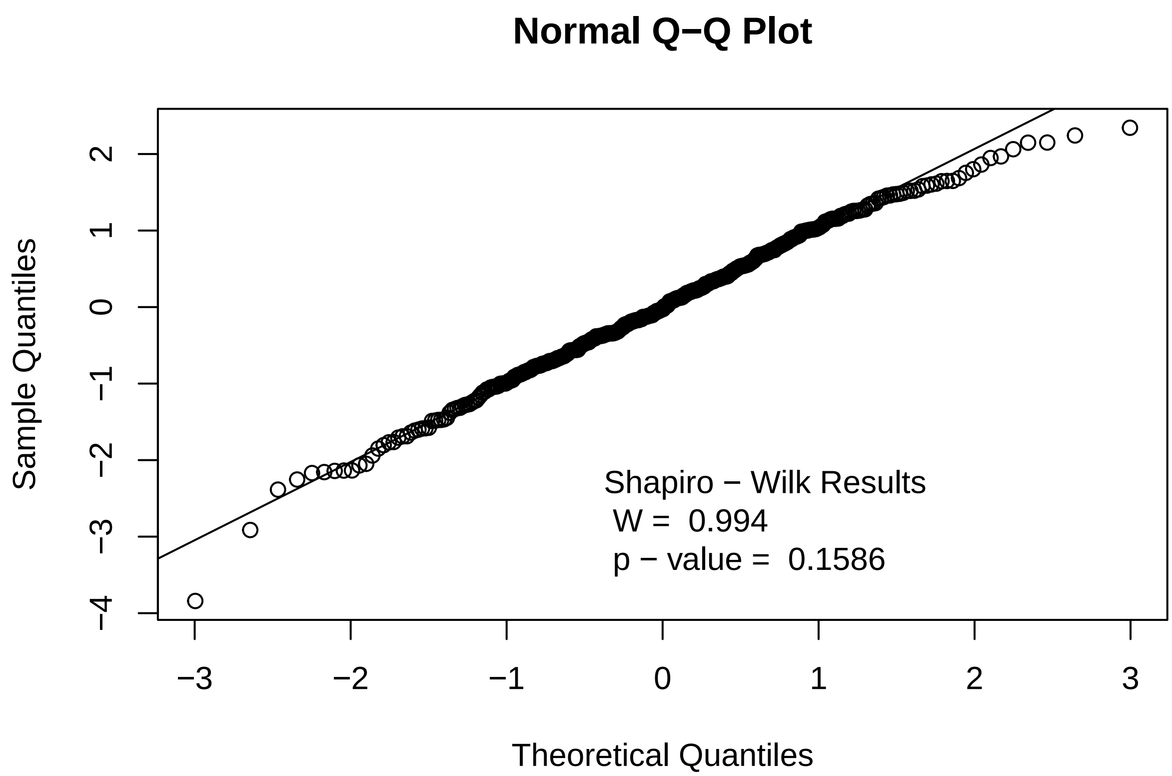

```
save.image()
```

Normality: Normal Q - Q plot with Shapiro - Wilk Test

```
#Normal Q - Q Plot
qqnorm(rstandard(lm5))
qqline(rstandard(lm5))

#Shapiro - Wilk Test
shap5 <- shapiro.test(resid(lm5))
```

```r
#Add Shapiro results to Normal Q - Q Plot
text(x = -0.5, y = -3,
     labels = paste("Shapiro - Wilk Results",
                    "\n W = ", round(shap5$statistic, 4),
                    "\n p - value = ", round(shap5$p.value, 4)),
      pos = 4, col = "black")
```



Conclusions

Final model for viewing.

```r
#Final Model
#dept and discipline removed by step()
lm5 <- lm((quality^3.55) ~ (gender + pepper
    + easiness + raterInterest + numYears
    + numCourses  + numRaters + sdQuality
    + sdHelpfulness + sdClarity + sdEasiness
    + sdRaterInterest)^2, Rateprof)
```

Table: Comparison of models used in the analysis.

The table includes: adjusted R^2, Shapiro - Wilk p - value, and Breusch - Pagan p - value.

The models: full additive model I, full stepwise model II, interactions model III, and final model IV.

```
# 1. Dynamically extract the p-values from your saved test objects
# (Assumes you saved your tests as shap1, bp1, shap2, bp2, etc. earlier in the script)

models_tb <- data.frame(
  "Model" = c("Full Additive I", "Full Step II", "Interactions III", "Final IV"),

  "Adj. $R^2$" = c(
    round(glance(lm1)$adj.r.squared, 4),
    round(glance(lm2)$adj.r.squared, 4),
    round(glance(lm3)$adj.r.squared, 4),
    round(glance(lm5)$adj.r.squared, 4)
  ),

  "Shapiro - Wilk" = c(
    format.pval(shap1$p.value, digits = 4),
    format.pval(shap2$p.value, digits = 4),
    format.pval(shap3$p.value, digits = 4),
    format.pval(shap5$p.value, digits = 4)
  ),

  "Breusch - Pagan" = c(
    format.pval(bp1$p.value, digits = 4),
    format.pval(bp2$p.value, digits = 4),
    format.pval(bp3$p.value, digits = 4),
    format.pval(bp5$p.value, digits = 4)
  ),

  check.names = FALSE
)

# 2. Render the comparison table using kable
kable(
  models_tb,
  caption = "Regression Models Comparison",
  align = c("l", "c", "c", "c"),
  format = "latex",
  booktabs = TRUE,
  escape = FALSE
)
```

Table 8: Regression Models Comparison

Model	Adj. R^2	Shapiro - Wilk	Breusch - Pagan
Full Additive I	0.9968	$< 2.2e-16$	0.3143
Full Step II	0.9972	$< 2.2e-16$	1.421e-08
Interactions III	0.6606	0.001737	0.3325
Final IV	0.8032	0.1586	0.1198